Test Series Code

09

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 22/03/2026

ANSWER KEY

1.	(2)	46.	(1)	91.	(4)	136.	(4)
2.	(3)	47.	(2)	92.	(2)	137.	(2)
3.	(2)	48.	(2)	93.	(3)	138.	(2)
4.	(3)	49.	(1)	94.	(4)	139.	(3)
5.	(2)	50.	(2)	95.	(1)	140.	(1)
6.	(1)	51.	(4)	96.	(3)	141.	(2)
7.	(1)	52.	(2)	97.	(2)	142.	(2)
8.	(2)	53.	(3)	98.	(1)	143.	(3)
9.	(3)	54.	(1)	99.	(3)	144.	(2)
10.	(1)	55.	(1)	100.	(4)	145.	(3)
11.	(1)	56.	(4)	101.	(4)	146.	(3)
12.	(3)	57.	(1)	102.	(3)	147.	(3)
13.	(3)	58.	(1)	103.	(1)	148.	(4)
14.	(3)	59.	(2)	104.	(2)	149.	(2)
15.	(3)	60.	(2)	105.	(2)	150.	(1)
16.	(2)	61.	(3)	106.	(3)	151.	(2)
17.	(1)	62.	(2)	107.	(1)	152.	(2)
18.	(1)	63.	(1)	108.	(3)	153.	(2)
19.	(1)	64.	(1)	109.	(4)	154.	(1)
20.	(3)	65.	(3)	110.	(3)	155.	(3)
21.	(2)	66.	(1)	111.	(4)	156.	(3)
22.	(2)	67.	(4)	112.	(1)	157.	(3)
23.	(1)	68.	(4)	113.	(2)	158.	(1)
24.	(1)	69.	(1)	114.	(1)	159.	(1)
25.	(1)	70.	(4)	115.	(2)	160.	(3)
26.	(1)	71.	(3)	116.	(2)	161.	(3)
27.	(2)	72.	(1)	117.	(4)	162.	(1)
28.	(3)	73.	(3)	118.	(3)	163.	(2)
29.	(3)	74.	(4)	119.	(1)	164.	(4)
30.	(1)	75.	(1)	120.	(4)	165.	(3)
31.	(2)	76.	(3)	121.	(3)	166.	(1)
32.	(3)	77.	(1)	122.	(2)	167.	(2)
33.	(1)	78.	(2)	123.	(1)	168.	(3)
34.	(1)	79.	(2)	124.	(4)	169.	(1)
35.	(3)	80.	(4)	125.	(1)	170.	(3)
36.	(2)	81.	(3)	126.	(2)	171.	(1)
37.	(4)	82.	(2)	127.	(3)	172.	(3)
38.	(2)	83.	(3)	128.	(2)	173.	(1)
39.	(3)	84.	(2)	129.	(2)	174.	(2)
40.	(1)	85.	(3)	130.	(1)	175.	(2)
41.	(1)	86.	(1)	131.	(1)	176.	(1)
42.	(2)	87.	(4)	132.	(4)	177.	(3)
43.	(1)	88.	(4)	133.	(3)	178.	(1)
44.	(1)	89.	(3)	134.	(4)	179.	(1)
45.	(1)	90.	(3)	135.	(1)	180.	(4)

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Hints & Solutions

1. (2)

Total Delay Time (t_0): The total time the student delays before running is the sum of preparing, drinking, and reacting.

$t_0 = 30\text{ s} + 18\text{ s} + 2\text{ s} = 50\text{ s}$

Equations of Motion: Let the train start at time $t = 0$ Distance covered by the train: $S_{\text{train}} = \frac{1}{2}at^2$

Distance covered by the student (who starts at $t = t_0$): $S_{\text{student}} = v(t - t_0)$

Equating Distances: To catch the train, their positions must match at some time t:

$\frac{1}{2}at^2 = v(t - t_0)$

$\frac{1}{2}at^2 - vt + vt_0 = 0$

Finding Minimum Speed: For the student to successfully catch the train, this quadratic equation must have real roots (discriminant $b^2 - 4ac \geq 0$).

$(-v)^2 - 4\left(\frac{a}{2}\right)(vt_0) \geq 0$

$v^2 - 2avt_0 \geq 0$

Dividing by v (since speed isn't zero): $v \geq 2at_0$

$v_{\min} = 2 \times (0.05\text{ m/s}^2) \times 50\text{ s}$

$v_{\min} = 5.0\text{ m/s}$

(NCERT Ed. 2025-26, 11th, Page No. 21)

2. (3)

The displacement current is uniformly distributed across the cross-section of the plates. Therefore, the displacement current passing through a specific area A' is proportional to that area:

$I_{d'} = i \times \left(\frac{A'}{A}\right)$

$I_{d'} = 4.0\text{ mA} \times \left(\frac{25\text{ cm}^2}{100\text{ cm}^2}\right) = 4.0 \times \frac{1}{4} = 1.0\text{ mA}$

(NCERT Ed. 2025-26, 12th, Page No. 202)

3. (2)

Net voltage $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

$= \sqrt{30^2 + (20 - 60)^2} = \sqrt{900 + 1600} = 50\text{ V}$

Power factor $\cos\phi = \frac{V_R}{V} = \frac{30}{50} = 0.6$

(NCERT Ed. 2025-26, 12th, Page No. 191)

4. (3)

$I_0 = \frac{V_0}{X_L} = \frac{V_0}{\omega L}$

If ω becomes 2ω and V_0 becomes $\frac{V_0}{2}$,

new current $I_0' = \frac{V_0/2}{2\omega L} = \frac{1}{4}\frac{V_0}{\omega L} = \frac{I_0}{4}$.

(NCERT Ed. 2025-26, 12th, Page No. 182)

5. (2)

$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = 100\ \Omega$

$\tan\phi = \frac{X_C}{R} = \frac{100}{100} = 1 \Rightarrow \phi = 45°$.

(NCERT Ed. 2025-26, 12th, Page No. 188)

6. (1)

The average energy density of the magnetic field is equal to that of the electric field:

$U_B = \frac{1}{4}\varepsilon_0 E_0^2 = 0.25 \times 8.85 \times 10^{-12} \times (60)^2$

$= 0.25 \times 8.85 \times 10^{-12} \times 3600$

$= 8.85 \times 10^{-12} \times 900 = 7.96 \times 10^{-9}\text{ J/m}^3$

(NCERT Ed. 2025-26, 12th, Page No. 209)

7. (1)

X-rays (A): Have very short wavelengths, typically around 1 nm (10^{-9} m). → (III)

Microwaves (B): Have wavelengths in the centimeter range, such as 1 cm (10^{-2} m). → (IV)

Visible light (C): Falls between 400 nm and 700 nm. 500 nm is a typical value. → (I)

AM Radio waves (D): Have long wavelengths, typically in the range of hundreds of meters. → (II)

(NCERT Ed. 2025-26, 12th, Page No. 211)

8. (2)

$\frac{1}{20} = (1.5 - 1)\left(\frac{2}{R}\right) \Rightarrow R = 20 \Rightarrow \text{cm}$.

In liquid: $\frac{1}{f_l} = \left(\frac{1.5}{1.6} - 1\right)\left(\frac{2}{20}\right)$

$= \left(\frac{-0.1}{1.6}\right)\left(\frac{1}{10}\right) = -\frac{1}{160}$

Thus, $f_l = -160\text{ cm}$ (Diverging).

(NCERT Ed. 2025-26, 12th, Page No. 234)

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9. (3)

$\theta =$ area under $\omega - t$ graph

$\theta = \frac{1}{2}(5)(30) + (15)(30) + \frac{1}{2}(5)(30)$

$\theta = 75 + 450 + 75 = 600\,\text{rad}$

(NCERT Ed. 2025-26, 11th, Page No. 117)

10. (1)

In normal adjustment (image at infinity),

Magnifying power $M = \frac{f_o}{f_e} = \frac{100}{5} = 20$

Tube length $L = f_o + f_e = 100 + 5 = 105\,\text{cm}$.

(NCERT Ed. 2025-26, 12th, Page No. 244)

11. (1)

The total apparent depth is the sum of the apparent depths through each medium:

$d_{\text{app}} = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} = \frac{12}{4/3} + \frac{3}{1.5} = 9 + 2 = 11\text{cm}$

(NCERT Ed. 2025-26, 12th, Page No. 229)

12. (3)

Each bounce reduces the height by a factor of e^2,

$h_n = e^{2n} \cdot h$

Given $e = \frac{1}{2}$, and n = 3:

$h_3 = (\frac{1}{2})^{2\times 3}$

$h_3 = \frac{1}{64}h$

13. (3)

$I_{res} = I + 4I + 2\sqrt{I \cdot 4I}\cos(60°)$

$I_{res} = 5I + 2(2I) \cdot \left(\frac{1}{2}\right)$

$I_{res} = 5I + 2I = 7I$

(NCERT Ed. 2025-26, 12th, Page No. 264)

14. (3)

Condition to be inside the Central Maximum: The dark fringe angle must be less than the diffraction minimum angle:

$(n - 0.5)\frac{\lambda}{d} < \frac{\lambda}{a}$

$n - 0.5 < \frac{d}{a}$

Substitute $d = 1.60$ mm and $a = 0.20$ mm:

$n - 0.5 < \frac{1.60}{0.20}$

$n - 0.5 < 8$

$n < 8.5$

The maximum integer value for n is 8.

This means there are 8 dark fringes on one side of the central maximum. By symmetry, there are another 8 on the other side.

Total dark fringes = 8 + 8 = 16

(NCERT Ed. 2025-26, 12th, Page No. 267)

15. (3)

After passing through P_1, the unpolarized light becomes polarized with intensity $I_1 = \frac{I_0}{2}$.

By Malus' Law, after P_2,

$I_2 = I_1\cos^2(60°) = \left(\frac{I_0}{2}\right) \times \left(\frac{1}{2}\right)^2 = \frac{I_0}{8}$.

(NCERT Ed. 2025-26, 12th, Page No. 270)

16. (2)

$\tan i_p = \mu$

$\tan i_p = \sqrt{3} \Rightarrow i_p = 60°$

At Brewster's angle, the reflected and refracted rays are perpendicular, meaning $i_p + r = 90°$

$r = 90° - 60° = 30°$

The deviation is the angle between the original path of the incident ray and its new refracted path.

$\delta = i_p - r = 60° - 30° = 30°$

17. (1)

Energy of incident photon

$E = \frac{hc}{\lambda} = \frac{1240}{200} = 6.2$ eV

From Einstein's photoelectric equation,

$K_{\max} = E - \Phi = 6.2 - 4.0 = 2.2$ eV

(NCERT Ed. 2025-26, 12th, Page No. 281)

18. (1)

de Broglie wavelength $\lambda = \frac{h}{\sqrt{2mK}}$

Since K is constant, $\lambda \propto \frac{1}{\sqrt{m}}$

Since $m_\alpha > m_p > m_e$, it follows that $\lambda_e > \lambda_p > \lambda_\alpha$

(NCERT Ed. 2025-26, 12th, Page No. 284)

19. (1)

A $\rightarrow$ II: Saturation current is proportional to the intensity (number of incident photons).

B $\rightarrow$ III: Stopping potential ($eV_0 = h\nu - \phi$) is determined by both incident frequency (ν) and the metal's work function (ϕ).

C $\rightarrow$ I: Work function is an intrinsic property of the specific material used.

(NCERT Ed. 2025-26, 12th, Page No. 278)

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20. (3)

$\omega = \dfrac{2\pi}{T}$

Time period $T = \dfrac{2\pi m}{qB}$ is independent of speed and mass is same in both cases.

(NCERT Ed. 2025-26, 12th, Page No. 112)

21. (2)

Energy of ground state $E_1 = -13.6$ eV.

Energy after absorption

$E_n = -13.6 + 12.09 = -1.51$ eV

We know

$E_n = \dfrac{-13.6}{n^2} \Rightarrow -1.51 = \dfrac{-13.6}{n^2} \Rightarrow n^2 = 9 \Rightarrow n = 3$

The max number of emission lines from $n = 3$ is

$\dfrac{n(n-1)}{2} = \dfrac{3 \times 2}{2} = 3$

(NCERT Ed. 2025-26, 12th, Page No. 300)

22. (2)

In a Bohr orbit, Kinetic Energy $K = -E$ (where E is Total Energy).

Thus, $\dfrac{K}{E} = -1$, meaning the ratio is $1:-1$.

(NCERT Ed. 2025-26, 12th, Page No. 296)

23. (1)

At the distance of closest approach,

Kinetic Energy = Electrostatic Potential Energy.

$K = \dfrac{1}{4\pi\varepsilon_0}\dfrac{(2e)(Ze)}{r_0}$

$5 \times 10^6 \times 1.6 \times 10^{-19} = \dfrac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{r_0}$

Solving, $r_0 = \dfrac{9 \times 10^9 \times 158 \times 1.6 \times 10^{-19}}{5 \times 10^6}$

$\approx 4.55 \times 10^{-14}$ m

(NCERT Ed. 2025-26, 12th, Page No. 295)

24. (1)

Statement I is true $\left(L = \dfrac{nh}{2\pi}\right)$.

Statement II is false; Rutherford said electrons go around the nucleus like planets, but he couldn't explain why they don't lose energy and crash into the nucleus.

(NCERT Ed. 2025-26, 12th, Page No. 298)

25. (1)

Nuclear radius $R \propto A^{1/3}$.

$\dfrac{R_2}{R_1} = \left(\dfrac{A_2}{A_1}\right)^{1/3} \Rightarrow \dfrac{R_2}{5} = \left(\dfrac{64}{125}\right)^{1/3} = \dfrac{4}{5}$

Thus, $R_2 = 4$ fermi

(NCERT Ed. 2025-26, 12th, Page No. 309)

26. (1)

Mass defect $\Delta m = 236.04 - 235.84 = 0.20$ u

Energy released $E = \Delta m \times 931.5$ MeV

$= 0.20 \times 931.5 = 186.3$ MeV

(NCERT Ed. 2025-26, 12th, Page No. 315)

27. (2)

For energy to be released (exothermic reaction), there must be a mass defect. The mass of the products must be strictly less than the mass of the reactants. Thus, $M_1 > M_2$.

(NCERT Ed. 2025-26, 12th, Page No. 315)

28. (3)

The binding energy per nucleon curve peaks near mass number $A = 56$. Iron (^{56}Fe) has the highest binding energy per nucleon (approx 8.8 MeV/nucleon), making it the most stable nucleus.

(NCERT Ed. 2025-26, 12th, Page No. 312)

29. (3)

Nuclear density is independent of mass number and remains ρ.

(NCERT Ed. 2025-26, 12th, Page No. 309)

30. (1)

The RMS voltage is $V_{rms} = 200$ V

The average power dissipated by the heater is

$P = \dfrac{V_{rms}^2}{R} = \dfrac{(200)^2}{100} = 400$ W $= 400$ J/s

Time $t = 4.2$ minutes $= 4.2 \times 60$ seconds

Total heat generated $H = 400 \times 4.2 \times 60$ joules

Heat required to melt m grams of ice is

$Q = m \times 80 \times 4.2$ joules

Equating H and Q: $400 \times 4.2 \times 60 = m \times 80 \times 4.2$

$\Rightarrow 400 \times 60 = m \times 80 \Rightarrow m = 300$ g

(NCERT Ed. 2025-26, 12th, Page No. 180)

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31. (2)

Let the vertically upward direction be positive (+).

Velocity of the moving mirror, $v_m = +2$ m/s.

The velocity of the dropped ball (object) after 1 sec,

$v_o = 0 + (-10) \times 1 = -10$ m/s (moving downwards).

For a plane mirror, the velocity of the image is given by the relation: $v_i = 2v_m - v_o$

$v_i = 2(+2) - (-10) = 4 + 10 = +14$ m/s

(moving upwards).

$v_{\text{rel}} = \mid v_i - v_o \mid = \mid 14 - (-10) \mid = 24$ m/s

32. (3)

Momentum Conservation: Initial momentum is zero. Therefore, $m_{\text{man}} v_{\text{man}} = m_{\text{woman}} v_{\text{woman}}$

$80 \times 2.5 = 50 \times v_{\text{woman}}$

$200 = 50 \times v_{\text{woman}} \Rightarrow v_{\text{woman}} = 4.0$ m/s.

(Statement 2 is correct, Statement 1 is incorrect).

Work-Energy Theorem:

Work done by the man = Kinetic energy gained by the woman =

$\frac{1}{2} m_{\text{woman}} v_{\text{woman}}^2 = \frac{1}{2} \times 50 \times (4.0)^2 = 25 \times 16 = 400$ J

(Statement 3 is correct).

Work done by the woman = Kinetic energy gained by the man

$= \frac{1}{2} m_{\text{man}} v_{\text{man}}^2 = \frac{1}{2} \times 80 \times (2.5)^2 = 40 \times 6.25 = 250$ J

Comparing Work: 250 J (done by woman) is less than 400 J (done by man). (Statement 4 is correct).

(NCERT Ed. 2025-26, 11th, Page No. 73)

33. (1)

$V_s = \frac{V_p}{10} = \frac{220}{10} = 22$ V

Secondary current $I_s = \frac{V_s}{R} = \frac{22}{22} = 1$A

For an ideal transformer,

$I_p V_p = I_s V_s \Rightarrow I_p = \frac{(I_s V_s)}{V_p} = \frac{(1 \times 22)}{220} = 0.1$A.

(NCERT Ed. 2025-26, 12th, Page No. 195)

34. (1)

Difference between any two consecutive frequency of Closed organ pipe

$= \frac{2v}{4l} = 260 - 220 = 40$ Hz

$\Rightarrow \frac{v}{4l} = 20$ Hz

So fundamental frequency = 20 Hz

(NCERT Ed. 2025-26, 11th, Page No. 292)

35. (3)

$E = \frac{-dV}{dx}$

$V(x) = ax^3 - bx + c$

$\frac{dV}{dx} = 3ax^2 - b$

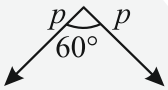
$E(x) = -(3ax^2 - b)$

$E(x) = -3ax^2 + b$

$E(x)$ is a quadratic function of the position x, Therefore, the electric field in this region varies parabolically with position.

(NCERT Ed. 2025-26, 12th, Page No. 55)

36. (2)



$p_{net} = \sqrt{3}p = \sqrt{3}(2qa)$, along $-y$-axis.

(NCERT Ed. 2025-26, 12th, Page No. 23)

37. (4)

Some metals, when cooled to very low temperatures, may exhibit both perfect conductivity and perfect diamagnetism.

The phenomenon of perfect diamagnetism in superconductors is called the Meissner effect.

The magnetic susceptibility χ and relative permeability μ_r for a perfect diamagnetic material are –1 and 0 respectively.

Superconducting magnets can be used in variety of situations like running magnetically levitated superfast trains.

(NCERT Ed. 2025-26, 12th, Page No. 148)

38. (2)

$\varepsilon = -M\left(\frac{di}{dt}\right)$

$\frac{d}{dt}[10\sin(100\pi t)] = 10 \times 100\pi \times \cos(100\pi t)$

$= 1000\pi \cos(100\pi t)$

The maximum value of $\cos(100\pi t)$ is 1, so the max rate of change is 1000π A/s.

Maximum EMF ($\varepsilon_{\max}$):

$M \times \left(\frac{di}{dt}\right)_{\max} = 0.005 \text{ H} \times 1000\pi \text{ A/s} = 5\pi$ V

(NCERT Ed. 2025-26, 12th, Page No. 166)

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39. (3)

Intensity I of a point source is inversely proportional to the square of the distance ($I \propto \frac{1}{r^2}$).

Halving r increases I by a factor of 4. Since saturation current is directly proportional to intensity, it becomes 4 times.

(NCERT Ed. 2025-26, 12th, Page No. 278)

40. (1)

In uniform circular motion, the acceleration is centripetal, directed towards the center.

$R = |\vec{r}| = \sqrt{(3^2 + 4^2)} = 5 \text{ m}$

$v = |\vec{v}| = \sqrt{(-8)^2 + 6^2} = \sqrt{(64+36)} = \sqrt{100} = 10 \text{ m/s}$

Magnitude of Acceleration (a_c):

$a_c = \frac{v^2}{R} = \frac{10^2}{5} = \frac{100}{5} = 20 \text{ m/s}^2$

Direction of Acceleration: The acceleration is directed opposite to the position vector (towards the center). The unit vector in the direction of $\vec{r}$ is

$\hat{r} = \frac{\vec{r}}{r} = \frac{(3\hat{i} + 4\hat{j})}{5}$

The acceleration vector $\vec{a}$ is a_c in the $-\hat{r}$ direction.

$\vec{a} = -a_c\hat{r} = -20 \times (3\hat{i} + 4\hat{j})/5$

$= -4 \times (3\hat{i} + 4\hat{j}) = -12\hat{i} - 16\hat{j} \text{ m/s}^2$

(NCERT Ed. 2025-26, 11th, Page No. 41)

41. (1)

Shift $y_0 = \frac{D}{d}(\mu - 1)t$

$= \left(\frac{1}{0.5 \times 10^{-3}}\right)(1.5 - 1)\left(1.5 \times 10^{-6}\right)$

$= 2000 \times 0.5 \times 1.5 \times 10^{-6} = 1.5 \times 10^{-3} \text{m} = 1.5 \text{mm}$

42. (2)

At surface: $V(R) = -\frac{GM}{R}$

At center: $V_c = -\frac{3GM}{2R}$

Given: $V(R) = -12 \text{J/kg} = -\frac{GM}{R}$

Thus: $V_c = \frac{3}{2}\left(-\frac{GM}{R}\right) = \frac{3}{2}(-12) = -18 \text{J/kg}$

43. (1)

Optical fibers do work on TIR (Statement I is true). TIR requires the angle of incidence to be strictly greater than the critical angle (Statement II is false).

(NCERT Ed. 2025-26, 12th, Page No. 231)

44. (1)

$\Delta U = -\Delta W = -\frac{R(T_1 - T_2)}{(\gamma - 1)} = \frac{R(T_2 - T_1)}{\gamma - 1}$

(NCERT Ed. 2025-26, 11th, Page No. 235)

45. (1)

With the voltmeter only, the voltage across the parallel group is $14.4 - 5.4 = 9.0 \text{ V}$.

Main current $I = \frac{9.0 \text{ V}}{10\,\Omega} = 0.9 \text{ A}$.

Equivalent resistance of R_3 & Voltmeter is $R_{3V} = \frac{5.4 \text{ V}}{0.9 \text{ A}} = 6\,\Omega$.

Since $R_3 = 12\,\Omega$ and $\frac{1}{R_V} + \frac{1}{12} = \frac{1}{6}$, we get $R_V = 12\,\Omega$.

With the ammeter only, let its resistance be R_A.

The voltage across the parallel part is $V_p = 0.24 \times (20 + R_A)$.

Total current $I = I_1 + I_2 = \frac{V_p}{20} + 0.24$.

Using Kirchhoff's voltage law: $V_p + I(12) = 14.4$.

Substituting V_p and I into the equation and solving gives $R_A = 10\,\Omega$.

Equivalent parallel resistance

$R_p = 20 \| (20 + 10) = 20 \| 30 = 12\,\Omega$.

Equivalent series resistance $R_{3V} = 12 \| 12 = 6\,\Omega$.

Total circuit resistance $= 12 + 6 = 18\,\Omega$.

Total current $I = \frac{14.4 \text{ V}}{18\,\Omega} = 0.8 \text{ A}$

$V = 0.8 \text{ A} \times 6\,\Omega = 4.8 \text{ V}$

Voltage across R_p is $0.8 \text{ A} \times 12\,\Omega = 9.6 \text{ V}$

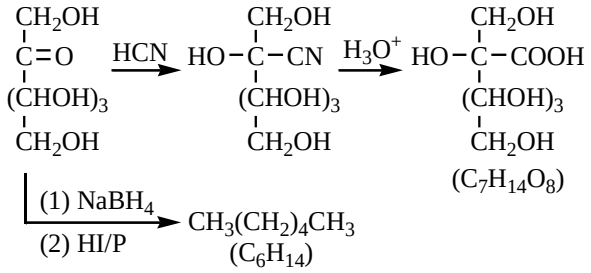
Current in ammeter branch

$= \frac{9.6 \text{ V}}{30\,\Omega} = 0.32 \text{ A} = 320 \text{ mA}$

(NCERT Ed. 2025-26, 12th, Page No. 97)

[Contd...

46. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 283)

47. (2)

$[Cr(H_2O)_5CO_3]Br$

pentaaquacarbonatochromium(III) bromide

(NCERT Ed. 2025-26 12th Part-I Page No. 124)

48. (2)

In 14th group, lead does not show catenation.

49. (1)

Oxidising power:

$MnO_4^- > Cr_2O_7^{2-} > VO_2^+$

(NCERT Ed. 2025-26 12th Part-I Page No. 100)

50. (2)

Volume of $O_2 = \frac{8}{32} \times 22.4\,L = 5.6\,L$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

51. (4)

- On hydrolysis proteins give rise to a large number of α-amino acids.
- Fibrous protein are generally water insoluble.

(NCERT Ed. 2025-26 12th Part-II Page No. 290)

52. (2)

Mole of $Cl^- = 0.1 \times 20 \times 10^{-3} \times 3 = 6 \times 10^{-3}$

Mole of Ag^+ required = $0.1 \times V \times 10^{-3}$

$\therefore\ 0.1 \times V \times 10^{-3} = 6 \times 10^{-3}$

V = 60 mL

(NCERT Ed. 2025-26 12th Part-I Page No. 120)

53. (3)

Thermodynamically most stable allotrope of carbon is graphite

54. (1)

- Scandium does not exhibit variable oxidation state.
- Cu^+ undergo disproportionation in aqueous solution.

(NCERT Ed. 2025-26 12th Part-I Page No. 96)

55. (1)

Isobars have same mass numbers

(NCERT Ed. 2025-26 11th Part-I Page No. 35)

56. (4)

Enzyme maltase catalysis the hydrolysis of maltose into glucose

(NCERT Ed. 2025-26 12th Part-II Page No. 295)

57. (1)

- $[Fe(C_2O_4)_3]^{3-}$ is more stable complex ion than $[Fe(H_2O)_6]^{3+}$ complex ion because in $C_2O_4^{2-}$ ligand chelation take place.
- Wilkinson catalyst is used for the hydrogenation of alkenes.

(NCERT Ed. 2025-26 12th Part-I Page No. 137)

58. (1)

- B_2O_3: acidic
- Al_2O_3, Ga_2O_3: amphoteric
- In_2O_3, Tl_2O_3: basic

59. (2)

Formation of complexes is due to:

(A) Comparatively smaller sizes of the metal ions

(B) High ionic charge on metal

(C) Availability of d-orbitals on metal

(NCERT Ed. 2025-26 12th Part-I Page No. 103)

60. (2)

For halogen, minimum oxidation state is –1.

(NCERT Ed. 2025-26 12th Part-II Page No. 240)

61. (3)

- Vitamin A: Xerophthalmia
- Thiamine: Beri-beri
- Ascorbic acid: Scurvy
- Riboflavin: Cheilosis

(NCERT Ed. 2025-26 12th Part-II Page No. 296)

[Contd...

62. (2)

Ligand field strength: $SCN^- < S^{2-} < NCS^-$

(NCERT Ed. 2025-26 12th Part-I Page No. 132)

63. (1)

Bond dissociation energy of E–H bonds:

$H_2O > H_2S > H_2Se > H_2Te$

64. (1)

- Wacker process: $PdCl_2$
- Dry battery cell: MnO_2
- Ziegler catalysts: $TiCl_4 + Al(CH_3)_3$
- 'Silver' UK coins: Cu/Ni alloy

(NCERT Ed. 2025-26 12th Part-I Page No. 114)

65. (3)

Mixture of petrol and diesel can be separated by fractional distillation

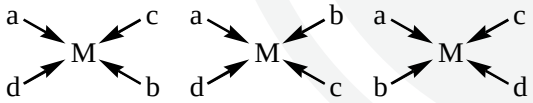
(NCERT Ed. 2025-26 11th Part-II Page No. 279)

66. (1)

Sucrose involve glycosidic linkage between C_1 of α-D-glucose and C_2 of β-D-fructose

(NCERT Ed. 2025-26 12th Part-II Page No. 287)

67. (4)



(NCERT Ed. 2025-26 12th Part-I Page No. 125)

68. (4)

BP: He < Ne < Ar < Kr < Xe

69. (1)

$\Delta_a H°$: V > Ni > Fe > Mn.

(NCERT Ed. 2025-26 12th Part-I Page No. 94)

70. (4)

$$\frac{p^0 - p_s}{p^0} = \frac{n}{n+N} \simeq \frac{n}{N}$$

$$\frac{50 - p_s}{50} = \frac{1.5}{\left(\frac{1000}{18}\right)}$$

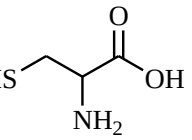
$$\therefore\ 50 - p_s = \frac{1.5 \times 50 \times 18}{1000} = 1.35$$

p_s = 48.65 mm of Hg

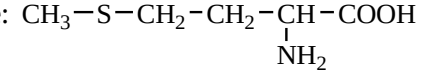
(NCERT Ed. 2025-26 12th Part-I Page No. 16)

71. (3)

Cysteine:



Methionine:

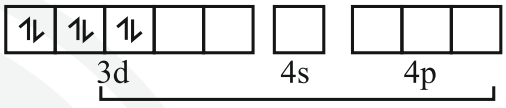


(NCERT Ed. 2025-26 12th Part-II Page No. 291)

72. (1)

Co^{3+} : $[Ar]3d^6 4s^0 4p^0 4d^0$

Co^{3+} in complex:



- d^2sp^3 hybridise
- Low spin
- Diamagnetic

(NCERT Ed. 2025-26 12th Part-I Page No. 130)

73. (3)

Among hydrogen halides, HF has highest boiling point because there is strong H-bonding between HF molecules.

74. (4)

Yb^{2+} which has f^{14} configuration is a reductant

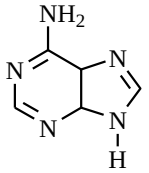
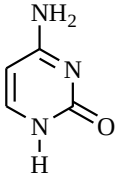
(NCERT Ed. 2025-26 12th Part-I Page No. 38)

75. (1)

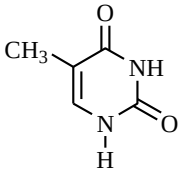
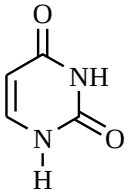
For concentration cell, where anode and cathode are of same element, $E^o_{cell} = 0$

(NCERT Ed. 2025-26 11th Part-II Page No. 00)

76. (3)

Adenine	
Cytosine	

[Contd...

Thymine	
Uracil	

(NCERT Ed. 2025-26 12th Part-II Page No. 298)

77. (1)

$cis\text{-}[CrCl_2(ox)_2]^{3-}$ is a chiral species

(NCERT Ed. 2025-26 12th Part-I Page No. 126)

78. (2)

CH_3CN contain both carbon and nitrogen.

(NCERT Ed. 2025-26 11th Part-II Page No. 284)

79. (2)

Iodonation of methane is very slow and reversible reaction

(NCERT Ed. 2025-26 11th Part-II Page No. 300)

80. (4)

Silica gel and alumina are used as adsorbent

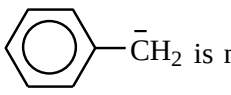
(NCERT Ed. 2025-26 11th Part-II Page No. 283)

81. (3)

Vitamin C are water soluble vitamins

(NCERT Ed. 2025-26 12th Part-II Page No. 296)

82. (2)

 is most stable due to presence of resonance and absence of +I effect.

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

83. (3)

meq. of H_2SO_4 = meq. of NH_3 + meq. of NaOH

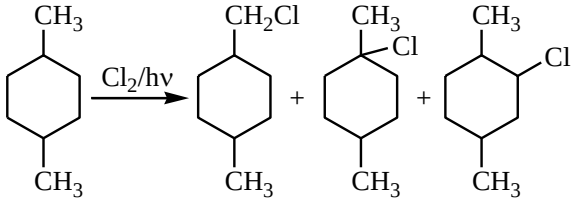
$$2\times\frac{1}{10}\times100=\text{meq. of } NH_3+200\times1\times\frac{1}{20}$$

$$\text{meq of } NH_3=10$$

$$\%\text{ of } N=\frac{\text{meq of } NH_3\times1.4}{\text{mass of compound}}=\frac{10\times1.4}{0.5}=28\%$$

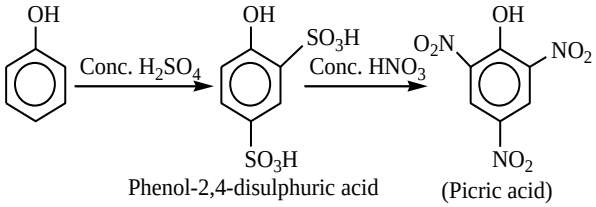
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84. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 164)

85. (3)

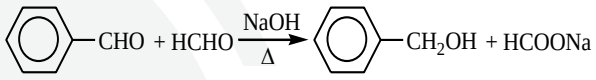


Phenol-2,4-disulphuric acid (Picric acid)

(NCERT Ed. 2025-26 12th Part-II Page No. 211)

86. (1)

Cross-Cannizaro reaction.



(NCERT Ed. 2025-26 12th Part-II Page No. 242)

87. (4)

Basic nature in vapour phase:

4 > 3 > 1 > 2

(NCERT Ed. 2025-26 12th Part-II Page No. 268)

88. (4)

The specific rate constant of a first order reaction depends on temperature and presence of catalyst

(NCERT Ed. 2025-26 12th Part-I Page No. 78)

89. (3)

	$A(g)\rightleftharpoons$	$B(g)$ +	$C(g)$
(*i*)	2 mol	–	–
(f)	1.5 mol	0.5 mol	0.5 mol

$$K_c=\frac{\left(\frac{0.5}{2}\right)\left(\frac{0.5}{2}\right)}{\left(\frac{1.5}{2}\right)}=0.083$$

(NCERT Ed. 2025-26 11th Part-I Page No. 183)

90. (3)

Methods used for purification of organic compounds are based on nature of compound and presence of impurity

(NCERT Ed. 2025-26 11th Part-II Page No. 278)

[Contd...

91. (4)
Humification leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate.
(NCERT Ed. 2025-26, 12th, Page No. 208)

92. (2)
When the number of reproductive individuals is more than that of pre- and post-reproductive individuals, the population is said to be declining.
(NCERT Ed. 2025-26, 12th, Page No. 191)

93. (3)
In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called apical dominance. Removal of shoot tips (decapitation) usually results in the growth of lateral buds. It is widely applied in tea plantations and hedge-making. Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in cotton, coriander and larkspur.
(NCERT Ed. 2025-26, 11th, Page No. 175)

94. (4)
In chemiosmosis, as electrons move through the photosystems, protons are transported across the membrane. This happens because the primary accepter of electron which is located towards the outer side of the membrane transfers its electron not to an electron carrier but to an H carrier (i.e. plastoquinone). Plastoquinone (PQ) is the oxidized form. When it accepts two electrons and two protons, it is reduced to plastoquinol (PQH_2).
(NCERT Ed. 2025-26, 11th, Page No. 141)

95. (1)
The three-domain system has also been proposed that divides the Kingdom Monera into two domains, leaving the remaining eukaryotic kingdoms in the third domain and thereby a six kingdom classification.
(NCERT Ed. 2025-26, 11th, Page No. 11)

96. (3)
First Trophic level — Tree
Second Trophic level — Grasshopper
Third Trophic level — Bird
Fourth Trophic level — Lion
(NCERT Ed. 2025-26, 12th, Page No. 210)

97. (2)
Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in cotton, coriander and larkspur.
(NCERT Ed. 2025-26, 11th, Page No. 173)

98. (1)
Although total number is generally the *most appropriate* measure of population density, it is in some cases either meaningless or difficult to determine. In an area, if there are 200 carrot grass (*Parthenium hysterophorus*) plants but only a single huge banyan tree with a large canopy, stating that the population density of banyan is low relative to that of carrot grass amounts to underestimating the enormous role of the Banyan in that community. In such cases, the per cent cover or biomass is a more meaningful measure of the population size.
(NCERT Ed. 2025-26, 12th, Page No. 192)

99. (3)
In C_4 plants photorespiration does not occur. This is because they have a mechanism that increases the concentration of CO_2 at the enzyme site. This takes place when the C_4 acid from the mesophyll is broken down in the bundle sheath cells to release CO_2 – this results in increasing the intracellular concentration of CO_2.
(NCERT Ed. 2025-26, 11th, Page No. 147)

100. (4)
Secondary productivity is defined as the rate of formation of new organic matter by consumers.
(NCERT Ed. 2025-26, 12th, Page No. 207)

101. (4)
Abscisic acid acts as a general plant growth inhibitor and an inhibitor of plant metabolism.
(NCERT Ed. 2025-26, 11th, Page No. 177)

102. (3)
The *lac* operon consists of one regulatory gene (the *i* gene) and three structural genes (*z*, *y*, and *a*). The *i* gene codes for the repressor of the *lac* operon. The *z* gene codes for beta-galactosidase (β-gal). The *y* gene codes for permease. The *a* gene encodes a transacetylase. Hence, all the three gene products in *lac* operon are required for metabolism of lactose.
(NCERT Ed. 2025-26, 12th, Page No. 100)

103. (1)
An orchid growing as an epiphyte on a mango branch, and barnacles growing on the back of a whale benefit while neither the mango tree nor the whale derives any apparent benefit. This is the interaction in which one species benefits and the other is neither harmed nor benefited. The association between fungi and the roots of higher plants is mutualism. The interaction between a tiger and a deer is predation. Cuckoo laying its eggs in a crow's nest represents brood parasitism.
(NCERT Ed. 2025-26, 12th, Page No. 201)

[Contd...

104. (2)

The organism shown in the diagram is a fungus. The detritus food chain (DFC) begins with dead organic matter. It is made up of decomposers which are heterotrophic organisms, mainly fungi and bacteria. Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as catabolism.

(NCERT Ed. 2025-26, 12th, Page No. 210)

105. (2)

Asexual reproduction in liverworts takes place by fragmentation of thalli, or by the formation of specialised structures called gemmae.

(NCERT Ed. 2025-26, 11th, Page No. 29)

106. (3)

The water splitting complex is associated with the PS II, which itself is physically located on the inner side of the membrane of the thylakoid. Since splitting of the water molecule takes place on the inner side of the membrane, the protons or hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids. According to the chemiosmotic hypothesis, ATP is synthesized by the enzyme ATP synthase. This enzyme has a portion (the F_1 headpiece) that protrudes from the thylakoid membrane into the stroma. As protons flow from the lumen to the stroma through the enzyme, ATP is generated on the stromal side of the thylakoid membrane.

(NCERT Ed. 2025-26, 11th, Page No. 140)

107. (1)

As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root. Pith is large and well developed. Monocotyledonous roots do not undergo any secondary growth. The vascular bundles have radial arrangement in roots.

(NCERT Ed. 2025-26, 11th, Page No. 74)

108. (3)

The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton. The pyramid of energy is always upright, can never be inverted because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.

(NCERT Ed. 2025-26, 12th, Page No. 213)

109. (4)

A population growing in a habitat with limited resources show initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote, when the population density reaches the carrying capacity. A plot of N in relation to time (t) results in a sigmoid curve. This type of population growth is called *Verhulst-Pearl Logistic Growth* and is described by the following equation:

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

(NCERT Ed. 2025-26, 12th, Page No. 195)

110. (3)

Ethylene breaks seed and bud dormancy, initiates germination in peanut seeds, sprouting of potato tubers.

(NCERT Ed. 2025-26, 11th, Page No. 177)

111. (4)

The photosynthetic pigments are organised into two discrete photochemical light harvesting complexes (LHC) within the Photosystem I (PS I) and Photosystem II (PS II). These are named in the sequence of their discovery, and not in the sequence in which they function during the light reaction.

(NCERT Ed. 2025-26, 11th, Page No. 138)

112. (1)

In marginal placentation the placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows, as in pea. When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose, tomato and lemon. In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part e.g., mustard and *Argemone*. When the ovules are borne on central axis and septa are absent, as in *Dianthus* and *Primrose* the placentation is called free central.

(NCERT Ed. 2025-26, 11th, Page No. 65)

113. (2)

Organism	Kingdom
Mycoplasma	Monera
Plasmodium	Protista
Ginkgo	Plantae
Neurospora	Fungi

(NCERT Ed. 2025-26, 11th, Page No. 14)

114. (1)

Ramdeo Misra is revered as the Father of Ecology in India.

(NCERT Ed. 2025-26, 12th, Page No. 188)

[Contd...

115. (2)
Gibberellins, cause fruits like apple to elongate and improve its shape.
(NCERT Ed. 2025-26, 11th, Page No. 176)

116. (2)
It is generally believed that competition occurs when closely related species compete for the same resources that are limiting, but this is not entirely true. Firstly, totally unrelated species could also compete for the same resource. Secondly, resources need not be limiting for competition to occur; in interference competition, the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species, even if resources (food and space) are abundant. In general, herbivores and plants appear to be more adversely affected by competition than carnivores.
(NCERT Ed. 2025-26, 12th, Page No. 198)

117. (4)
Light Reaction: Oxygen, along with the formation of ATP and NADPH.
Glycolysis: Pyruvate, along with the formation of ATP and NADH.
Fermentation in yeast: Ethanol and carbon dioxide.
Photorespiration: Phosphoglycolate and Phosphoglycerate.
(NCERT Ed. 2025-26, 11th, Page No. 156)

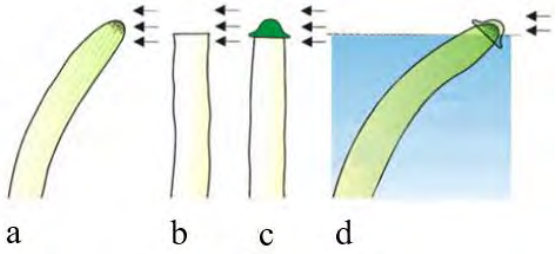
118. (3)
If fatty acids were to be respired they would first be degraded to acetyl CoA and enter the pathway. But when the organism needs to synthesise fatty acids, acetyl CoA would be withdrawn from the respiratory pathway for it. Acetyl CoA is a 2-carbon compound. It enters the Krebs' cycle during aerobic respiration. PEP is the primary acceptor of CO_2 in C_4 plants.
(NCERT Ed. 2025-26, 11th, Page No. 162)

119. (1)
Duplication of DNA: during the S-phase of interphase.
Pairing of homologous chromosomes: prophase I.
Formation of dyad of cells: cytokinesis after telophase I.
Splitting of centromere: anaphase II.
Formation of 4 haploid cells: cytokinesis after telophase II.
(NCERT Ed. 2025-26, 11th, Page No. 126)

120. (4)
Except for the deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on Earth. The flow of energy from the sun to producers and then to consumers is unidirectional.
(NCERT Ed. 2025-26, 12th, Page No. 209)

121. (3)
The red algae usually reproduce vegetatively by fragmentation. They reproduce asexually by non-motile spores and sexually by non-motile gametes. Sexual reproduction is oogamous and accompanied by complex post-fertilisation developments.
(NCERT Ed. 2025-26, 11th, Page No. 28)

122. (2)
Charles Darwin and his son Francis Darwin observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism). After a series of experiments, it was concluded that the tip of coleoptile was the site of transmittable influence that caused the bending of the entire coleoptile.



Experiment used to demonstrate that tip of the coleoptile is the source of auxin. Arrows indicate direction of light
(NCERT Ed. 2025-26, 11th, Page No. 174)

123. (1)
Following double fertilisation, events of endosperm development from the central cell of embryo sac, embryo development from zygote, maturation of ovule(s) into seed(s) and ovary into fruit, are collectively termed post-fertilisation events. Integuments of ovules harden as tough protective seed coats and the antipodals degenerate.
(NCERT Ed. 2025-26, 12th, Page No. 20)

124. (4)

Organism	Kingdom	Phylum	Class	Order
Human	Animalia	Chordata	Mammalia	Primata
Monkey	Animalia	Chordata	Mammalia	Primata
Cat	Animalia	Chordata	Mammalia	Carnivora
Housefly	Animalia	Arthropoda	Insecta	Diptera

(NCERT Ed. 2025-26, 11th, Page No. 8)

125. (1)
The two DNA chains have anti-parallel polarity. It means, if one chain has the polarity 5'→3', the other has 3'→5'. The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). A purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix.
(NCERT Ed. 2025-26, 12th, Page No. 81)

[Contd...

126. (2)

Pond ecosystem: The autotrophic components include the phytoplankton, some algae and the floating, submerged and marginal plants found at the edges. The consumers are represented by the zooplankton, the free swimming and bottom dwelling forms. The decomposers are the fungi, bacteria and flagellates especially abundant in the bottom of the pond.

(NCERT Ed. 2025-26, 12th, Page No. 206)

127. (3)

Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings. E. Kurosawa (1926) reported the appearance of symptoms of the 'bakanae' disease in rice seedlings when they were treated with sterile filtrates of the fungus. The active substances were later identified as gibberellic acid. Miller et al. (1955) identified and crystallised the cytokinesis promoting active substance that they termed kinetin. H.H. Cousins (1910) confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas. Later this volatile substance was identified as ethylene.

(NCERT Ed. 2025-26, 11th, Page No. 174)

128. (2)

The photochemical reactions, since they are directly light driven are called light reactions. In stroma, enzymatic reactions synthesise sugar, which in turn forms starch. These are not directly light driven but are dependent on the products of light reactions (ATP and NADPH). Hence, to distinguish the latter they are called, by convention, as dark reactions (carbon reactions). However, this should not be construed to mean that they occur in darkness or that they are not light-dependent. The dark reactions being enzymatic are temperature controlled.

(NCERT Ed. 2025-26, 11th, Page No. 136)

129. (2)

If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food. This is the reason why predators in nature are 'prudent'. Besides acting as 'conduits' for energy transfer across trophic levels, predators play other important roles.

(NCERT Ed. 2025-26, 12th, Page No. 198)

130. (1)

Each trophic level has a certain mass of living material at a particular time called as the standing crop. The rate of biomass production is called productivity. Vertical distribution of different species occupying different levels is called stratification. Decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition.

(NCERT Ed. 2025-26, 12th, Page No. 211)

131. (1)

A centriole is made up of nine evenly spaced peripheral fibrils of tubulin protein. Each of the peripheral fibril is a triplet. Therefore, total number of microtubules = $9 \times 3 = 27$.

(NCERT Ed. 2025-26, 11th, Page No. 100)

132. (4)

In the hypogynous flower the gynoecium occupies the highest position while the other parts are situated below it. The ovary in such flowers is said to be superior, e.g., mustard, china rose and brinjal. If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous. The ovary here is said to be half-inferior, e.g., plum, rose, peach.

(NCERT Ed. 2025-26, 11th, Page No. 63)

133. (3)

Calvin proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances.

(NCERT Ed. 2025-26, 11th, Page No. 132)

134. (4)

A chromatographic separation of the leaf pigments shows that the colour that we see in leaves is not due to a single pigment but due to four pigments: Chlorophyll *a* (bright or blue green in the chromatogram), chlorophyll *b* (yellow green), xanthophylls (yellow) and carotenoids (yellow to yellow-orange).

(NCERT Ed. 2025-26, 11th, Page No. 137)

135. (1)

The cells derived from root apical and shoot-apical meristems and cambium differentiate and mature to perform specific functions. This act leading to maturation is termed as differentiation. During differentiation, cells undergo few to major structural changes both in their cell walls and protoplasm. For example, to form a tracheary element, the cells would lose their protoplasm. They also develop a very strong, elastic, lignocellulosic secondary cell walls.

(NCERT Ed. 2025-26, 11th, Page No. 172)

136. (4)

DNA can enter cells if the membrane is artificially made permeable. During chemical treatment with divalent cations (e.g., $CaCl_2$) followed by heat shock, bacterial cells become competent and allow DNA to pass through the membrane.

DNA is highly hydrophilic and negatively charged, whereas the core of the plasma membrane is hydrophobic (lipid bilayer). Because of this polarity mismatch, DNA cannot naturally diffuse across the membrane.

(NCERT Ed. 2025-26, 12th, Page No. 170)

137. (2)

The rop (repressor of primer) gene encodes a protein that regulates plasmid replication and helps maintain the plasmid copy number in pBR322. Hence it is involved in replication control.

The *BamH*I site lies within the tetracycline resistance (tet^R) gene, not the ampicillin resistance gene. Insertion of foreign DNA at *BamH*I therefore inactivates tetracycline resistance, not ampicillin resistance.

The ampicillin resistance gene (amp^R) in pBR322 contains restriction sites such as *Pst*I and *Pvu*I, which can be used for insertional inactivation.

Selectable markers (amp^R or tet^R) are used to select transformants from non-transformants, not to differentiate recombinant from non-recombinant transformants. Differentiation is done by insertional inactivation of the marker.

(NCERT Ed. 2025-26, 12th, Page No. 169)

138. (2)

The water vascular system is a characteristic feature of phylum Echinodermata (e.g., starfish, sea urchins). It is a hydraulic system consisting of madreporite, stone canal, ring canal, radial canals, and tube feet.

Functions include:

- Locomotion (via tube feet)
- Capture and transport of food
- Respiration
- Excretion

(NCERT Ed. 2025-26, 11th, Page No. 45)

139. (3)

In recombinant DNA technology, both vector DNA and foreign DNA must be cut with the same restriction enzyme so that they produce complementary sticky ends.

If X (restriction enzyme) is different for vector and foreign DNA, the sticky ends produced will not be complementary, so they cannot base-pair.

Because of this:

The fragments cannot anneal properly.

DNA ligase cannot efficiently join them.

Therefore, recombinant DNA will not form.

(NCERT Ed. 2025-26, 12th, Page No. 168)

140. (1)

List-I	List-II
P-wave	Depolarisation of the atria
QRS complex	Depolarisation of the ventricles
T-wave	Repolarisation of the ventricles
Cardiac output	Volume of blood pumped out by each ventricle per minute

(NCERT Ed. 2025-26, 11th, Page No. 201)

141. (2)

Juxtaglomerular (JG) cells are modified smooth muscle cells located in the afferent arteriole of the nephron. They are activated when:

- Blood pressure falls
- Sodium concentration in distal tubule decreases
- Sympathetic stimulation increases

Immediate consequence:

JG cells secrete renin into circulation.

Renin acts on angiotensinogen (produced by liver) and converts it into angiotensin I, which is later converted to angiotensin II by ACE.

(NCERT Ed. 2025-26, 11th, Page No. 212)

142. (2)

Contraction is initiated by a motor neuron, not a sensory neuron.

Acetylcholine binds to receptors, causing depolarisation of sarcolemma.

Ca^{2+} binds to troponin, causing conformational change and exposing myosin-binding sites.

ATP hydrolysis energizes myosin head enabling cross-bridge formation.

During contraction, Z lines move towards each other, shortening the sarcomere.

(NCERT Ed. 2025-26, 11th, Page No. 222)

143. (3)

The cranial meninges consist of an outer layer called dura mater, a very thin middle layer called arachnoid, and an inner layer which is in contact with the brain tissue called pia mater. The corpus callosum is a tract of nerve fibres connecting the two cerebral hemispheres, not a meningeal layer.

(NCERT Ed. 2025-26, 11th, Page No. 235)

144. (2)

List-I	List-II
Thyrocalcitonin	Decreases blood Ca^{2+} levels
Parathyroid hormone	Increases blood Ca^{2+} levels
Insulin	Hypoglycemic hormone
Aldosterone	Main mineralocorticoid in the human body

(NCERT Ed. 2025-26, 11th, Page No. 243)

[Contd...

145. (3)

Oogenesis begins during embryonic development. In the fetal ovary, millions of oogonia are formed and enter meiosis I to become primary oocytes.

Each ovary is covered by a thin germinal epithelium, not a thick epithelium. Beneath it lies the ovarian stroma.

(NCERT Ed. 2025-26, 12th, Page No. 32)

146. (3)

According to the Medical Termination of Pregnancy (Amendment) Act, 2017, a pregnancy may be terminated within the first 12 weeks on the opinion of one registered medical practitioner. If the pregnancy has lasted more than 12 weeks but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required grounds exist.

(NCERT Ed. 2025-26, 12th, Page No. 46)

147. (3)

Convergent evolution occurs when unrelated organisms independently evolve similar traits due to adaptation to similar environmental pressures. These structures are called analogous organs.

Eye of octopus and eye of mammals → Developed independently in molluscs and vertebrates but perform the same function → Classic example of convergent evolution.

(NCERT Ed. 2025-26, 12th, Page No. 115)

148. (4)

Microinjection involves directly injecting recombinant DNA into the nucleus of an animal cell using a fine glass micropipette.

Also called the gene gun method, where DNA-coated gold or tungsten micro-particles are bombarded into plant cells at high velocity.

A virus or bacterium is genetically modified so it cannot cause disease, but it can still deliver foreign DNA into host cells.

Ascariasis is caused by Ascaris lumbricoides.

In bacterial transformation, cells are first treated with ice-cold $CaCl_2$ to make them competent and then briefly exposed to ~42°C heat shock, which facilitates DNA entry. 100°C would kill the cells,

(NCERT Ed. 2025-26, 12th, Page No. 171)

149. (2)

When foreign DNA is inserted into the beta-galactosidase gene, the enzyme is inactivated (insertional inactivation). On a chromogenic substrate, bacteria without the insert produce blue colonies because the enzyme is active. Bacteria with the recombinant plasmid cannot produce the active enzyme and thus their colonies do not produce any colour, marking them as recombinants.

(NCERT Ed. 2025-26, 12th, Page No. 170)

150. (1)

In recombinant DNA technology, a foreign DNA fragment cannot replicate independently inside a host cell unless it is linked to a DNA molecule that possesses an Origin of Replication (ori).

For multiplication of an alien DNA, it must be inserted into a vector or chromosome containing an ori. The replication machinery of the host recognizes this site and begins replication, allowing the inserted DNA to multiply along with the vector.

The origin of replication is the specific DNA sequence where replication begins. It also controls the copy number of the linked DNA fragment or plasmid inside the host cell.

(NCERT Ed. 2025-26, 12th, Page No. 164)

151. (2)

The cloning vector pBR322 contains two selectable marker genes:

- amp^R → ampicillin resistance
- tet^R → tetracycline resistance

In pBR322, the *Sal* I restriction site lies within the tetracycline resistance (tet^R) gene.

When foreign DNA is inserted at the *Sal* I site, it causes insertional inactivation of the tet^R gene.

Consequences:

- amp^R gene remains functional → bacteria remain ampicillin resistant.
- tet^R gene is disrupted → bacteria become tetracycline sensitive.

Therefore, transformants with recombinant plasmid:

- Grow on ampicillin medium
- Do not grow on tetracycline medium

(NCERT Ed. 2025-26, 12th, Page No. 169)

152. (2)

Nitrogenous bases (heterocyclic rings) attached to a sugar are called nucleosides. Adenosine, guanosine, thymidine, uridine, and cytidine are nucleosides. Adenylic acid and guanylic acid are nucleotides (containing a phosphate group). Adenine and Cytosine are nitrogenous bases alone.

(NCERT Ed. 2025-26, 11th, Page No. 106)

153. (2)

The head of cockroach is formed by fusion of six segments. It is connected to the thorax by a flexible cervical membrane (neck), which allows movement in almost all directions.

Tegmina are the forewings of cockroach. They are dark, leathery, and opaque. At rest, they cover and protect the delicate membranous hind wings used for flight.

In both males and females, the abdomen fundamentally has 10 segments. However, in females, the 7th–9th segments are reduced or concealed due to modification into genital structures. Thus, the number is not 9 in males.

In males, the genital pouch is located at the posterior end of the abdomen and is bounded dorsally by the 9th and 10th terga and ventrally by the 9th sternum.

Males possess a pair of short, thread-like anal styles on the 9th sternum, which are absent in females.

(NCERT Ed. 2020-21, 11th, Page No. 112)

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154. (1)

In biotechnology, Downstream Processing refers to the steps involved after the formation of the desired product in bioreactors.

Downstream processing includes:

- Separation of the product from the culture medium
- Purification of the product
- Formulation and quality testing before marketing

These steps ensure the product reaches the required purity and safety standards.

(NCERT Ed. 2025-26, 12th, Page No. 174)

155. (3)

In PCR, DNA amplification occurs in three steps:

1. Denaturation (≈94–95°C) – Double-stranded DNA separates into single strands.
2. Annealing (≈50–65°C) – Primers bind (hybridize) to complementary sequences on the template DNA.
3. Extension (≈72°C) – DNA synthesis occurs using *Taq* DNA polymerase.

During the annealing step, two primers are essential:

- Forward primer
- Reverse primer

These primers attach to opposite strands of the template DNA and define the region that will be amplified.

(NCERT Ed. 2025-26, 12th, Page No. 172)

156. (3)

For efficient cloning using a vector such as pBR322, certain essential features are required.

Origin of replication (ori) – Allows the vector and inserted DNA to replicate inside the host cell.

Selectable marker – Helps identify transformants (e.g., antibiotic resistance genes).

Cloning sites (restriction sites) – Unique restriction enzyme sites where foreign DNA can be inserted.However, multiple recognition sites for the same restriction enzyme are undesirable. If a vector had several sites for the same enzyme, digestion would produce multiple fragments, making cloning difficult.

Therefore, vectors usually have a single (unique) restriction site for each enzyme.

(NCERT Ed. 2025-26, 12th, Page No. 169)

157. (3)

DNA ligase acts as the "molecular glue" that forms phosphodiester bonds to join DNA fragments.

(NCERT Ed. 2025-26, 12th, Page No. 167)

158. (1)

The bacterium *Agrobacterium tumefaciens* naturally contains the Ti plasmid, which causes crown gall tumors in plants.

In biotechnology, this plasmid is modified (disarmed) by removing the tumor-causing genes and is then used as a cloning vector to transfer foreign DNA into plant cells. Hence it becomes non-pathogenic.

Certain Retrovirus can transform normal animal cells into cancerous cells because their DNA integrates into the host genome.

In genetic engineering, these viruses are disarmed (disease-causing genes removed) and used as vectors to deliver foreign genes into animal cells.

(NCERT Ed. 2025-26, 12th, Page No. 170)

159. (1)

In Agarose Gel Electrophoresis, DNA fragments are separated based on size when an electric field is applied.

- DNA is negatively charged due to its phosphate backbone.
- It moves towards the positive electrode (anode).
- Smaller DNA fragments move faster and travel farther through the agarose gel pores.
- 100 bp DNA fragment → smallest → travels farthest
- 200 bp DNA fragment → larger → travels less distance
- 1000 bp DNA fragment → much larger → moves slower
- Circular plasmid (5000 bp) → very large and circular → moves the least

(NCERT Ed. 2025-26, 12th, Page No. 168)

[Contd...

160. (3)

The father produces two types of sperms — X-bearing and Y-bearing — whereas the mother produces only X-bearing ova.

Male gametes (sperms) → carry either X or Y

Female gametes (ova) → carry only X

(NCERT Ed. 2025-26, 12th, Page No. 36)

161. (3)

Vigorous stirring often creates foam which can interfere with oxygen transfer and lead to contamination. The foam breaker is positioned at the top of the liquid level to dissipate it.

(NCERT Ed. 2025-26, 12th, Page No. 174)

162. (1)

About 2000 million years ago (mya) the first cellular forms of life appeared on earth.

The archaic human species *Homo neanderthalensis* had an average brain size of about 1400 cc and lived roughly between 100,000–40,000 years ago in regions of Europe, the Near East, and parts of Central Asia.

(NCERT Ed. 2025-26, 12th, Page No. 121)

163. (2)

The symptoms described (sustained high fever, weakness, stomach pain, constipation, loss of appetite) are classic for typhoid fever, caused by the pathogenic bacterium *Salmonella typhi*. Typhoid fever is confirmed medically using the Widal test.

(NCERT Ed. 2025-26, 12th, Page No. 131)

164. (4)

List-I	List-II
Gel electrophoresis	Separating DNA fragments based on their size under an electric field.
Polymerase Chain Reaction (PCR)	Synthesizing multiple copies of the gene of interest *in vitro.*
Salmonella typhimurium	Construction of first rDNA molecule.
Plasmid	Autonomously replicating circular extra-chromosomal DNA)

(NCERT Ed. 2025-26, 12th, Page No. 168)

165. (3)

Specific bacteria gain resistance to antibiotics due to changes in their plasmids. Plasmids can carry genes that confer resistance to antibiotics, and these can be transferred between bacteria, spreading resistance.

(NCERT Ed. 2025-26, 12th, Page No. 164)

166. (1)

In enzyme naming (e.g., *EcoR*I), E is Genus, co is species, R is strain, and I (Roman numeral) is the order in which the enzyme was isolated from that strain.

(NCERT Ed. 2025-26, 12th, Page No. 166)

167. (2)

Almost all enzymes are proteins, but there are some exceptions where nucleic acids behave like enzymes. These are called ribozymes. They are RNA molecules with catalytic power, demonstrating that not all biological catalysts are proteins.

(NCERT Ed. 2025-26, 11th, Page No. 112)

168. (3)

In molecular biology, a palindrome refers to a sequence of double-stranded DNA where the reading direction (e.g., 5' to 3') is the same on both strands. Example: 5' - GAATTC - 3'

3' - CTTAAG - 5' If you read both in the 5' → 3' direction, they are identical.

(NCERT Ed. 2025-26, 12th, Page No. 167)

169. (1)

When a gene from one species is expressed in a different species (the host), the resulting protein is termed a recombinant protein.

(NCERT Ed. 2025-26, 12th, Page No. 173)

170. (3)

Squamous epithelium consists of a single thin layer of flattened cells, ideal for diffusion boundaries in blood vessels and alveoli.

(NCERT Ed. 2025-26, 11th, Page No. 101)

171. (1)

List-I	List-II
T-DNA	Transform normal plant cells into a tumor
Bioreactor	Large scale production
Vector	Plasmid or Virus
Agarose	Natural polymer extracted from sea weeds

(NCERT Ed. 2025-26, 12th, Page No. 169)

172. (3)

Taq DNA polymerase is not used in the denaturation step. Denaturation occurs simply due to high temperature, which separates DNA strands.

Taq polymerase is a thermostable DNA polymerase isolated from the bacterium *Thermus aquaticus*. Its heat stability allows it to remain active during repeated PCR cycles.

(NCERT Ed. 2025-26, 12th, Page No. 173)

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173. (1)

In continuous culture, fresh medium is added and used medium is drained to keep cells in their physiologically most active exponential phase.

(NCERT Ed. 2025-26, 12th, Page No. 174)

174. (2)

Ethidium bromide intercalates into DNA and fluoresces bright orange under UV light.

(NCERT Ed. 2025-26, 12th, Page No. 168)

175. (2)

List-I	List-II
EFB	European Federation of Biotechnology
Cloning	Making multiple identical copies of any template DNA
Bioprocess engineering	Maintenance of sterile (microbial contamination-free) ambience
Sexual reproduction	Provides opportunities for variations

(NCERT Ed. 2025-26, 12th, Page No. 166)

176. (1)

A probe is not a cloning vector rather it is a single-stranded sequence of DNA or RNA used to search for its complementary sequence in a sample genome.

(NCERT Ed. 2025-26, 12th, Page No. 183)

177. (3)

During DNA extraction, the ethanol can be gently pushed down into the DNA solution by using a glass rod or wooden spooling stick. The charged DNA molecules move away from the ethanol and can be scooped up and swirled around a glass rod or wooden spooling stick. This is called DNA spooling. DNA spooling can isolate long strands of DNA from a solution.

(NCERT Ed. 2025-26, 12th, Page No. 171)

178. (1)

DNA is insoluble in chilled ethanol, so when cold ethanol is added, DNA precipitates as fine white threads.

The first recombinant DNA was constructed in 1972 by Stanley Cohen and Herbert Boyer, marking the beginning of recombinant DNA technology.

(NCERT Ed. 2025-26, 12th, Page No. 164)

179. (1)

Recombinant DNA technology is commonly called Genetic engineering, as it involves purposeful alteration of DNA.

Recombinant DNA technology includes cutting, modifying, and joining DNA fragments and transferring the engineered DNA into a suitable host organism for expression of desired traits.

(NCERT Ed. 2025-26, 12th, Page No. 163)

180. (4)

Recognition sequence of first restriction endonuclease Hind II is of 6 base pairs and it generates blunt ends.

(NCERT Ed. 2025-26, 12th, Page No. 165)

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